

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The limited availability of expert annotations in specialized domains such as medical imaging constitutes a fundamental obstacle to training robust vision backbones under supervised objectives. Self-supervised masked image modeling (MIM) mitigates this constraint by defining a pretext reconstruction task on unlabeled data; however, prevailing formulations sample the mask uniformly at random under a high masking ratio, which induces both sample inefficiency and weak alignment between the pretext signal and downstream semantics. Region-aware variants partially address this deficiency, but condition mask placement on reconstruction-error heuristics or task-specific supervision, thereby constraining transferability across tasks and imaging modalities. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The framework employs a frozen vision-language model to perform prompt-conditioned region localization, and subsequently assigns spatially differentiated masking ratios that concentrate the reconstruction objective on diagnostically salient regions while attenuating redundancy over background. This controllable formulation tightens the coupling between the pretraining signal and downstream objectives, improving representation quality and cross-task transfer. We evaluate the framework across three imaging modalities—brain MRI, chest CT, and lung X-ray. Relative to established MIM baselines (e.g., SparK), Mask What Matters yields consistent improvements of up to +3.1 percentage points in top-1 classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection and instance segmentation. Critically, these gains are attained under a substantially reduced overall masking ratio (40% versus 70%). These results establish that controllable, text-conditioned masking provides a semantically aligned pretext objective for self-supervised learning, and advance the construction of robust vision backbones for medical image analysis.

1 INTRODUCTION

Image classification, semantic segmentation, and object detection constitute the canonical tasks of computer vision. Over the past decade, deep neural networks have become the dominant technical substrate for these tasks, achieving state-of-the-art performance under the empirical risk minimization paradigm. This performance, however, is contingent on access to large-scale annotated corpora. In specialized domains such as medical image processing, expert-verified annotations are both scarce and costly to acquire, which imposes a fundamental barrier to the deployment of supervised deep learning methods.

Self-supervised learning (SSL) circumvents this dependence by defining pretext objectives over unlabeled data, thereby inducing transferable feature representations without manual annotation (Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)). Within the SSL taxonomy, MIM has become a prominent paradigm. Formally, MIM corrupts an input image by masking a subset of its patches and trains an encoder-decoder to reconstruct the masked content conditioned on the visible context, which compels the model to internalize the structural and semantic regularities of the data distribution. This objective transfers effectively to a broad range

of downstream tasks and has been adapted to the medical imaging domain [He et al., (2022); Gupta et al., (2025); Xiao et al., (2023)].

Nevertheless, prevailing MIM instantiations—notably masked autoencoders (MAE) [He et al., (2022)] and SparK [Tian et al., (2023)]—sample the mask from a spatially uniform distribution that is independent of the regions of interest relevant to downstream tasks. This task-agnostic sampling incurs two principal limitations. (1) The induced representations are prone to semantic misalignment with the downstream objective, which degrades transfer performance. (2) To raise the probability of occluding informative content under uniform sampling, existing methods adopt very high masking ratios (e.g., 75% in MAE); this inflates the difficulty of the reconstruction task and, in turn, the data and compute required for convergence. Natural language, by contrast, exhibits compact structure and high semantic density, and thus supports effective masked pretraining at low masking ratios [Devlin et al., (2019)]. The dependence on high masking ratios in image-domain MIM is largely attributable to the pronounced spatial redundancy of natural images and the absence of semantic-aware mask placement, which together prevent the model from concentrating capacity on genuinely informative regions.

This limitation is especially acute in medical imaging, which is characterized by high semantic sparsity: diagnostically relevant information is typically confined to small, localized structures (e.g., lesions or organs), whereas the majority of the spatial extent is semantically redundant background. Moreover, the domain spans heterogeneous modalities—including MRI, CT, and X-ray—with substantial inter-task variation in the shape, scale, and spatial distribution of task-relevant regions. These properties impose correspondingly stringent requirements on the efficiency and cross-task generalizability of the masking strategy.

To address these limitations, we introduce a controllable, text-guided masking framework, hereafter denoted **Mask-What-Matters (MWM)**. Conditioned on a user-specified, open-vocabulary description or phrase, a pretrained vision–language model supplies zero-shot localization cues, which we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—assigning higher ratios to semantically salient regions (e.g., lesions and organs) and lower ratios to background—thereby injecting downstream semantics directly into the pretext objective without recourse to per-image reports or labels.

To summarize, our main contributions are:

- *A controllable text-guided masking framework for medical imaging.* To our knowledge, MWM is the first such framework: it composes a vision–language model (BiomedCLIP) with a segmentation-based refinement stage (SAM) to localize task-relevant regions from open-vocabulary prompts, and then imposes differentiated ROI-versus-background masking ratios, thereby eliminating the semantic misalignment inherent to uniform random masking.
- *An annotation-free and backbone-agnostic design.* MWM requires neither per-image reports nor labels, and integrates into both ViT- and ConvNet-based MIM pipelines under a single unified protocol.
- *Consistent gains across modalities and tasks.* On brain MRI, chest CT, and lung X-ray datasets, MWM surpasses state-of-the-art MIM baselines (e.g., SparK), improving classification, detection, and segmentation metrics and substantiating the utility of text-conditioned generative pretraining in medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first formalized in natural language processing by the Bidirectional Encoder Representations from Transformers (BERT) model [Devlin et al., (2019)], which learns contextual token representations by masking a subset of input tokens and predicting them from bidirectional context. Building on this formulation, [He et al., (2022)] transferred the masked-prediction objective to the visual domain, masking spatial patches during pretraining and requiring the model to reconstruct the occluded content. The paradigm exhibits strong transfer in both natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative instantiations include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian

et al. (2023). MAE couples a high-ratio uniform mask with an asymmetric encoder–decoder to bias ViTs toward high-level semantic abstraction; MedMAE specializes this objective to medical images through domain-specific adaptations; and SparK reformulates the masking scheme for convolutional backbones, exploiting their intrinsic multi-scale feature hierarchy to strengthen pretraining.

Notwithstanding these advances, conventional MIM instantiations sample masks from a random or structure-agnostic distribution that is decoupled from the semantic requirements of downstream tasks. Such non-selective sampling can allocate reconstruction capacity to task-irrelevant regions during pretraining, attenuating transfer effectiveness. To mitigate this, recent work has investigated region-aware masking distributions that improve semantic alignment, along two principal directions.

The first direction formulates loss-driven adaptive masking. Methods such as hard patches mining (HPM) Wang et al. (2023a) and its medical extension AnatoMask Li et al. (2024) estimate per-region reconstruction error online and assign higher masking probability to high-error regions, on the premise that reconstruction difficulty correlates with semantic value. However, this coupling can steer the objective toward regions that are hard to reconstruct yet task-irrelevant. In brain magnetic resonance imaging (MRI), for instance, high-frequency structures such as the spinal cord or eyeballs may attract disproportionate masking mass despite being unrelated to downstream targets such as tumor detection.

The second direction conditions mask placement on an external perceptual module. FocusMAE Basu et al. (2024), for example, employs a pretrained Region Proposal Network (RPN) to nominate high-information regions for selective masking. Although effective on specific tasks, this design requires supervised training of the RPN on domain-specific annotated data, which curtails its cross-task generalization.

Collectively, these limitations motivate a masking mechanism that is simultaneously flexible, semantically aligned, and generalizable. Natural language—a rich, interpretable, and readily available supervisory signal—constitutes a promising basis for guiding masked modeling in a task-aware yet annotation-free manner.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Advances in vision–language alignment have rendered accurate, zero-shot localization of task-relevant regions increasingly tractable under open-world settings Radford et al. (2021). The Contrastive Language–Image Pretraining (CLIP) family induces a shared image–text embedding space via large-scale contrastive pretraining on paired data, and has been adopted broadly for zero-shot and open-set visual recognition. In the medical domain, specialized variants such as BiomedCLIP Zhang et al. (2023) recalibrate this cross-modal alignment to biomedical distributions, attaining robust performance across heterogeneous downstream applications. Although vision–language models are now standard components in supervised pipelines, existing studies predominantly target downstream applications such as object detection and semantic segmentation (Zhong et al. 2022; Aleem et al. 2024; Koleilat et al. 2024), and have not examined whether their open-vocabulary, zero-shot localization capacity can be repurposed to supervise the masking distribution in masked image modeling.

In summary, extant region-aware MIM methods—whether based on reconstruction-loss-driven adaptive masking or on external perception modules—improve semantic focus only partially, and remain constrained by their reliance on auxiliary supervision, residual misalignment between masked regions and task semantics, and limited flexibility. Concurrently, notwithstanding the strong downstream performance of vision–language models, natural-language prompts have seen little use as a source of task-aware masking guidance during the MIM pretraining phase. These observations jointly motivate our approach: injecting vision–language semantic guidance into the pretext objective to yield efficient and generalizable masked image modeling.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As depicted in Figure 1, **Mask What Matters (MWM)** decomposes into two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage employs a

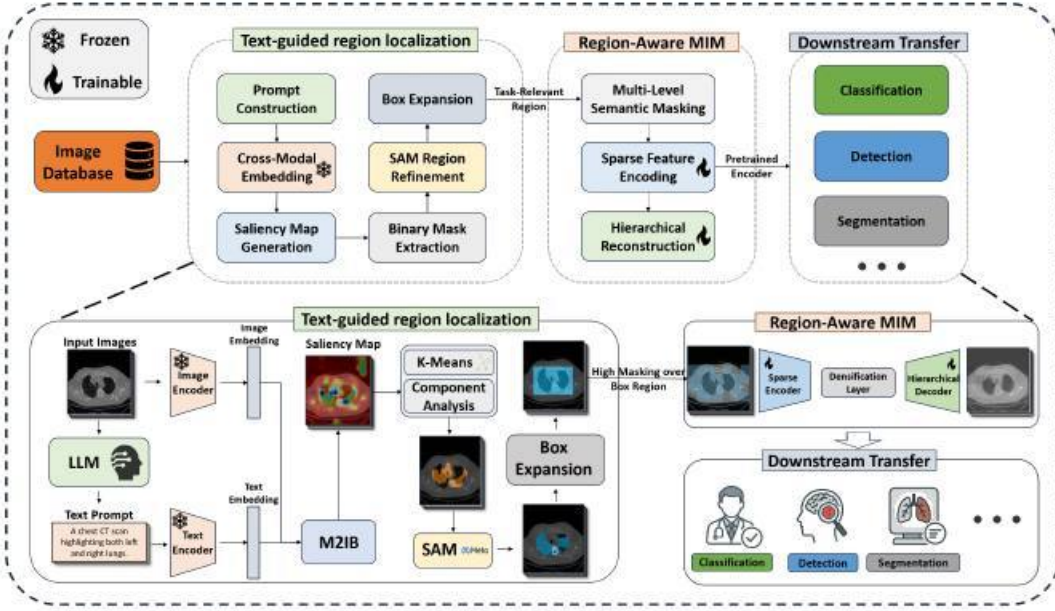


Figure 1: Architecture of the MWM framework. The top panel depicts the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel provides a detailed view of the mapping from textual prompts to localized regions and the resulting masking distribution.

frozen vision–language model to infer task-relevant regions from an open-vocabulary prompt; the second stage conditions the reconstruction objective on these regions by imposing spatially differentiated masking ratios.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a multi-stage pipeline. The pipeline first generates a textual prompt via an LLM and encodes the image–text pair with BiomedCLIP to obtain cross-modal embeddings. A saliency map is then estimated through the Multi-Modal Information Bottleneck (M2IB), binarized by K-Means clustering, and spatially refined by the Segment Anything Model (SAM). Finally, the refined mask is converted into an axis-aligned bounding box with a controllable expansion margin, which furnishes a robust spatial prior for the subsequent masked modeling stage.

Prompt Generation. In medical image analysis, the specification of the textual prompt materially governs the fidelity of region localization. To strengthen semantic guidance, we query large language models (LLMs) with a structured template of the form “Describe the typical visual characteristics of a [category] in a [modality] image”. As an alternative to such full-sentence prompts, we additionally evaluate bare category phrases as inputs. A controlled comparison of the two prompt styles is reported in the experimental section.

Cross-Modal Embedding. Given the generated prompt, we apply a BiomedCLIP model [Zhang et al., (2023)] to project the medical image and its prompt into a shared semantic embedding space. For an image I and a prompt T , the embeddings are given by

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings constitute the semantic basis for the saliency estimation performed in the subsequent stage.

Saliency Map Generation. To localize task-relevant visual regions, we adopt a cross-modal saliency estimator based on the Multi-modal Information Bottleneck (M2IB) [Wang et al., (2023b)], which aligns the image and text embeddings while suppressing modality-specific redundancy. Concretely, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns to each spatial location an importance weight

proportional to its semantic relevance to the prompt, obtained by optimizing the objective

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings produced by BiomedCLIP, and the coefficient γ governs the trade-off between retaining cross-modal relevance and discarding task-irrelevant visual information.

Binary Mask Extraction. We binarize the saliency map via unsupervised K-Means clustering [Lloyd (1982)] to delineate the prompt-conditioned region, yielding a preliminary foreground mask. The map is partitioned into two pixel clusters, and the cluster with the higher mean saliency is designated foreground. To suppress spurious activations, we apply connected-component analysis to the binary map and retain only the largest foreground component, which discards noisy boundary responses and concentrates the mask on the core semantic structure.

Region Refinement. To improve spatial fidelity, we incorporate SAM as a refinement module [Kirillov et al. (2023)]. For each retained connected component $c_i \in \mathcal{C}^*$, we compute its minimal enclosing bounding box and supply it as the visual prompt to SAM.

Conditioned on the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} yields a spatially more faithful delineation of the target region.

Box-Based Region Expansion. The SAM mask produced above retains residual boundary noise, which can bias the model’s attention during representation learning. To render the region guidance robust to such noise, we convert the refined SAM mask into an axis-aligned bounding box with a controllable expansion margin; this box constitutes the final region of interest (ROI) supplied to the downstream masked modeling stage.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest produced by the localization module of Section 3.2, MWM performs masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that assigns differentiated masking ratios as a function of region importance; (2) a sparse encoder that processes only the visible patches; and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM instantiates a multi-level masking distribution parameterized by the semantic regions returned by the text-driven localization stage. Patches falling within prompt-highlighted regions—e.g., those associated with tumors or organs—are masked at a high ratio, concentrating the reconstruction objective on task-relevant semantic content. Conversely, background patches are masked at a low ratio or left visible, which curtails redundant computation. Figure 2 contrasts the resulting masking patterns with uniform random masking.

Sparse Feature Encoding. The visible patches are aggregated into a sparse image and encoded following the SparK formulation [Tian et al. (2023)]:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i denotes the i -th patch, $m_i \in \{0, 1\}$ its visibility indicator, Φ_{sparse} the sparse encoder, and $F = \{F_1, F_2, F_3, F_4\}$ the resulting set of multi-scale feature maps. Although we instantiate MWM with convolutional backbones, the same interface transfers to ViTs through token indexing of the visible set.

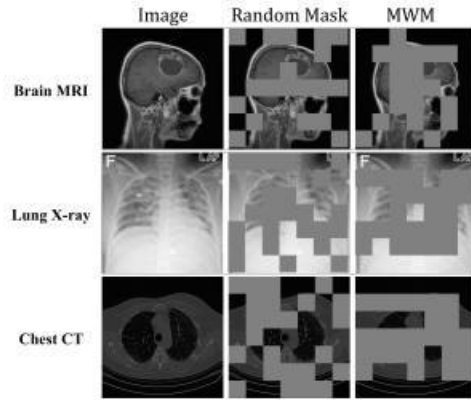


Figure 2: Comparison of masking strategies. Gray blocks denote masked patches.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down, coarse-to-fine manner. At each scale l , masked locations are populated with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified feature map F'_l is projected by a scale-specific layer ϕ_l to yield D_l , which is then upsampled and additively fused with the features at the next-lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is optimized by the mean-squared reconstruction error computed over the masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ denotes the index set of masked patches, x_i the original patch, and $\hat{x}_i$ its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining is conducted on three datasets spanning distinct imaging modalities. (1) *Brain MRI* aggregates approximately 17,000 images drawn from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], and covers a range of tumor types and anatomical structures. (2) *Lung X-ray* comprises approximately 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; [Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* contains 12,000 slices from a lung disease dataset [Konya (2020)], accompanied by fibrotic-lung segmentation masks for 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation over classification and detection tasks uses three datasets. (1) *Brain Tumor MRI* [Feltrin (2023)] provides 4,479 MRI images spanning 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermany et al. (2018)] contains 5,863 pediatric chest radiographs labeled as normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] comprises CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

To ensure comparability, all self-supervised methods use backbones of comparable parameter scale: convolution-based methods adopt ResNet-50 [He et al. (2016)], and transformer-based methods adopt ViT-S [Dosovitskiy et al. (2020)]. The input resolution is fixed at 224×224 , and all backbones

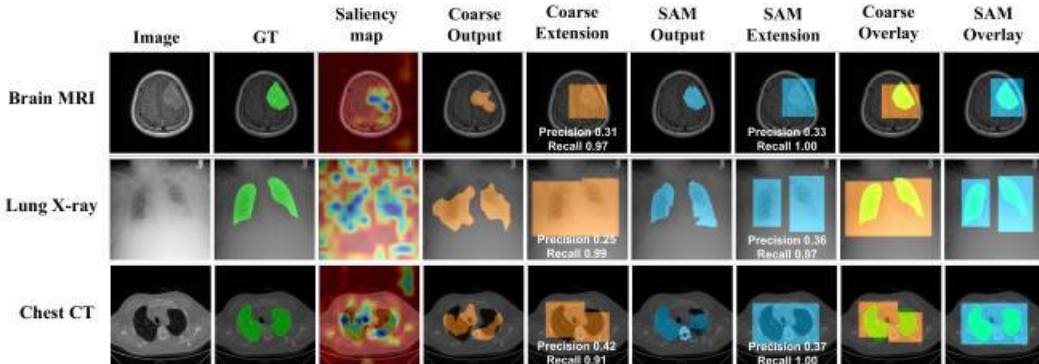


Figure 3: Representative qualitative outputs of the text-guided region localization stage.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (occlusion precision / recall) as a function of prompt type and refinement strategy. *Coarse Ext.* denotes bounding-box expansion of the coarse region; + *SAM Ext.* denotes additional SAM-based refinement prior to expansion.

are initialized from ImageNet-pretrained weights. Pretraining employs an initial learning rate of 2×10^{-4} . Downstream fine-tuning preserves the pretrained encoder and the input resolution. All experiments are executed on NVIDIA Tesla A100 GPUs with 80 GB of VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless stated otherwise, non-target factors (e.g., prompt type and SAM usage) are held at their empirically optimal settings to enforce controlled comparison.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first quantify the effectiveness of the text-guided localization module across three imaging modalities—brain MRI, lung X-ray, and chest CT—and then isolate the contributions of prompt type and SAM refinement. Predicted bounding boxes are compared against expert-annotated segmentation masks under two set-overlap metrics: occlusion precision, $|R_p \cap R_{gt}| / |R_p|$, and occlusion recall, $|R_p \cap R_{gt}| / |R_{gt}|$, where R_p denotes the predicted region and R_{gt} the ground-truth annotation.

As reported in Table 1, text-guided localization under effective prompts attains consistently high occlusion recall across all datasets (approximately 0.82) at reasonable precision. This confirms that the framework recovers task-relevant regions from textual descriptions alone, notwithstanding modality-specific anatomical variation; representative qualitative examples are shown in Figure 3.

Effect of Prompt Type on Region Localization. To characterize the influence of prompt design, we contrast the two prompt types. As Table 1 indicates, descriptive-sentence prompts are superior on brain MRI and chest CT, consistent with richer semantics improving localization. On lung radiographs, however, concise phrases yield higher recall (0.82 versus 0.66), plausibly because the broad anatomical extent of the lungs renders general terms sufficient for accurate localization.

Effect of SAM Refinement on Region Localization. We further quantify the contribution of spatial refinement by applying SAM to the coarse masks. As shown in Table 1, SAM refinement raises precision while preserving high recall: on brain MRI, precision increases from 0.22 to 0.30 with recall held above 0.94. This substantiates the role of SAM in sharpening coarse localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next assess the generalization and representational capacity of MWM against representative self-supervised learning (SSL) baselines drawn from both the reconstruction-based and contrastive paradigms. To ensure a controlled comparison, all methods are pretrained and evaluated under identical datasets and training configurations.

Image Classification. We evaluate classification on three downstream medical imaging datasets under two standard transfer protocols: (1) *full fine-tuning*, in which all backbone parameters are updated; and (2) *linear probing*, in which the encoder is frozen and only a linear classification head is trained on top of its features.

- **Full fine-tuning:** Table 2 reports full-fine-tuning accuracy for MWM alongside representative self-supervised baselines: MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and BYOL [Grill et al. (2020)].

MWM attains the highest accuracy on every modality, evidencing robust cross-modality generalization. The largest margin arises on chest CT, where MWM reaches 97.5% accuracy, exceeding SparK by +3.1 and AnatoMask by +1.7 points. This margin is consistent with the design rationale of text-guided masking, which concentrates the reconstruction objective on semantically critical yet spatially sparse structures—such as lung lobes in CT—while suppressing redundant masking of background.

- **Linear probing:** To probe representation quality independently of backbone adaptation, we conduct linear-probing experiments and, for concision, compare MWM against the two strongest recent MIM baselines, SparK and AnatoMask. As Table 3 shows, MWM outperforms both across all datasets, corroborating that text-driven pretraining induces more linearly separable features.

Object Detection and Instance Segmentation. To assess transfer beyond classification, we evaluate brain-tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As Table 4 reports, MWM attains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), exceeding SparK by +1.3 and +1.1 points, respectively. These results indicate that the representations learned by MWM transfer effectively to both object-level localization and pixel-level segmentation.

In aggregate, injecting text-driven masking into the pretraining objective yields more informative and transferable representations. This semantic guidance produces consistent gains across classification, detection, and segmentation, corroborating its efficacy for visual representation learning.

4.5 ABLATION STUDY

We conduct ablations that isolate the individual contributions of prompt design, SAM-based region refinement, and the masking ratio to downstream classification performance in MWM.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As reported in Table 5 descriptive-sentence prompts outperform short phrases on brain MRI and chest CT, whereas the ordering inverts on lung X-ray. This pattern mirrors the localization results of Section 4.3: prompts that yield more accurate region identification also yield stronger downstream accuracy. Crucially, both prompt types—irrespective of granularity—dominate uniform random masking on every dataset. This confirms that text-guided masking induces semantic alignment between the pretext and downstream objectives, a property central to the observed gains in self-supervised performance. Furthermore, the SAM ablation shows that enabling refinement during pretraining consistently improves downstream classification, corroborating its contribution to text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4 MWM attains 96.8% classification accuracy at a masking ratio of only 40%, substantially below the $\approx 70\%$ conventionally employed in

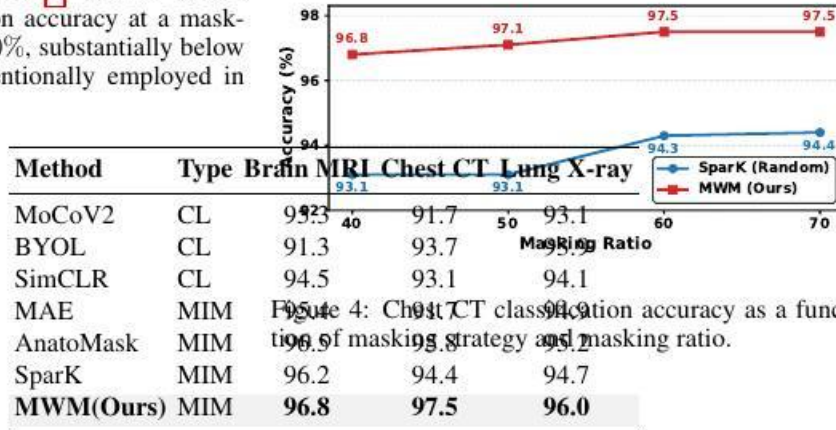


Table 2: Full fine-tuning classification accuracy (%) on the downstream datasets. MIM and CL denote Masked Image Modeling and Contrastive Learning, respectively.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear-probing classification accuracy (%) on the downstream datasets under a frozen encoder.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Object detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) performance (%) on the BR35H dataset.

masked image modeling. Despite this reduced ratio, MWM exceeds SparK’s best configuration at 70% masking by +2.4 points in absolute accuracy. To preclude the confound that SparK’s lower accuracy stems from a suboptimal masking configuration, we sweep SparK over masking ratios of 40%, 50%, 60%, and 70%. The sweep confirms that 70% masking is optimal for SparK, which validates the fairness of the comparison. These findings contradict the prevailing assumption that high masking ratios are intrinsically optimal for masked image modeling. Instead, they demonstrate that semantically guided, task-aware masking, as instantiated by MWM, achieves stronger downstream performance at lower masking ratios by preserving task-relevant content and eliminating redundancy.

5 CONCLUSION AND LIMITATIONS

This work introduces **Mask What Matters (MWM)**, a self-supervised pretraining framework that couples text-guided semantic localization with region-aware masking. By using natural-language prompts to localize task-relevant regions and imposing spatially differentiated masking ratios, MWM realizes semantically aligned representation learning without per-image annotations.

Our experiments establish the effectiveness and generalizability of MWM along three axes: (1) natural-language prompts reliably guide semantic region localization, validating the feasibility of text-driven masking; (2) MWM consistently surpasses existing methods (e.g., SparK, AnatoMask) in classification across three imaging modalities, and further improves detection and instance-segmentation metrics; (3) MWM sustains strong performance even at reduced masking ratios (e.g., 40%), quantifying the benefit of semantic guidance in self-supervised pretraining.

Limitations. Two limitations qualify these results. First, although textual prompts afford flexible guidance, we have not systematically characterized the sensitivity of MWM to variation in prompt style, vocabulary, or noise. Second, the dependence on externally supplied textual descriptions introduces a coupling that may degrade generalization in settings where reliable or consistent prompts are unavailable.

Looking forward, we contend that text-driven, region-aware masking constitutes a principled route toward more semantically grounded self-supervised learning, and that extending MWM beyond medical imaging may enable broader vision–language pretraining.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) as a function of prompt type and region-refinement strategy across the three imaging modalities. *No Prompt* corresponds to uniform random masking without text-guided localization. *w/* and *w/o* denote the presence and absence of SAM refinement, respectively.

REFERENCES

- Sana Aleem, Feiran Wang, Midhun Maniparambil, Eric Arazo, Johannes Dietlmeier, Kevin Curran, Krzysztof J. Geras, Sotirios A. Tsaftaris, Simon J. D. Prince, William M. Wells, William Lotter, Seung Wook Kim, Sebastian Pölsterl, and Samuel Little. Test-time adaptation with salip: A cascade of sam and clip for zero-shot medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5184–5193, 2024.
- Sourya Basu, Manan Gupta, Chahat Madan, Pulkit Gupta, and Chetan Arora. Focusmae: Gallbladder cancer detection from ultrasound videos with focused masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11715–11725, 2024.
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pp. 1597–1607. PMLR, 2020a. URL <https://proceedings.mlr.press/v119/chen20j.html>
- Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020b.
- Jin Cheng, Wei Huang, Shuangliang Cao, Riqiang Yang, Wenjia Yang, Zhaoqiang Yun, and et al. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLOS ONE*, 10(10):e0140381, 2015.
- Md. E. H. Chowdhury, Tanzila Rahman, Ahsan Khandakar, Rashid Mazhar, Md. Abdul Kadir, Zaid B. Mahbub, and et al. Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access*, 8:132665–132676, 2020.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, volume 1, pp. 4171–4186, 2019.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Ali Fateh, Yalda Rezvani, Sadegh Moayedi, Sam Rezvani, Farzaneh Fateh, and Mohammad Fateh. Brisc: Annotated dataset for brain tumor segmentation and classification with swin-hafnet. *arXiv preprint arXiv:2506.14318*, 2025.
- Fernando Feltrin. Brain tumor mri images 44 classes, 2023.
- Jean-Bastien Grill, Florian Strub, Florent Alché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, and et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 21271–21284, 2020.

- Amit Gupta, Islam Osman, Mohamed S. Shehata, William J. Braun, and Richard E. Feldman. Medmae: A self-supervised backbone for medical imaging tasks. *Computation*, 13(4):88, 2025.
- Ahmed Hamada. Br35h :: Brain tumor detection 2020, 2025.
- Mohamed Hany. Chest ct-scan images dataset for lung cancer classification, 2020.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009, 2022.
- Stanley C. Huang, Abhishek Pareek, Michael Jensen, Matthew P. Lungren, Serena Yeung, and Akshay S. Chaudhari. Self-supervised learning for medical image classification: A systematic review and implementation guidelines. *NPJ Digital Medicine*, 6(1):74, 2023.
- Anukriti Jaiswal, Abhinav Rajendra Babu, Mojtaba Zadeh Zadeh, Debanjan Banerjee, and Fillia Makedon. A survey on contrastive self-supervised learning. *Technologies*, 9(1):2, 2020.
- Daniel Kermany, Kang Zhang, and Michael Goldbaum. Labeled optical coherence tomography (oct) and chest x-ray images for classification. Mendeley Data, v2, 2018. URL <https://doi.org/10.17632/rschbjbr9sj.2>.
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, and et al. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4015–4026, 2023.
- Tareq Koleilat, Hamed Asgariandehkordi, Hassan Rivaz, and Yuyin Xiao. Medclip-sam: Bridging text and image towards universal medical image segmentation. In *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 643–653, Cham, 2024. Springer Nature Switzerland.
- Sandor Konya. Ct lung & heart & trachea segmentation, 2020.
- Rahul Krishnan, Pranav Rajpurkar, and Eric J. Topol. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering*, 6(12):1346–1352, 2022.
- Yufei Li, Tianyi Luan, Yuhang Wu, Shuhao Pan, Ying Chen, and Xujie Yang. Anatomask: Enhancing medical image segmentation with reconstruction-guided self-masking. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 146–163, Cham, 2024. Springer Nature Switzerland.
- Xin Liu, Fan Zhang, Zhenyong Hou, Lingqiao Mian, Zi Huang, Jian Zhang, and Jinhui Tang. Self-supervised learning: Generative or contrastive. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 35(1):857–876, 2021.
- Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*, 28(2): 129–137, 1982.
- Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, and et al. The multimodal brain tumor image segmentation benchmark (brats). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2014.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pp. 8748–8763. PMLR, 2021.

- Tanzila Rahman, Ahsan Khandakar, Yazan Qiblawey, Abdulla Tahir, Serkan Kiranyaz, Sheikh Abul Kashem, Md. Islam, Somaya Al Maadeed, Samar Zughaier, M. A. Khan, and Md. E. H. Chowdhury. Exploring the effect of image enhancement techniques on COVID-19 detection using chest x-rays images. *Computers in Biology and Medicine*, pp. 104319, 2021. doi: 10.1016/j.combiomed.2021.104319.
- Kai Tian, Yifan Jiang, Qihang Diao, Chaojian Lin, Li Wang, and Zehuan Yuan. Designing BERT for convolutional networks: Sparse and hierarchical masked modeling. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- Haoran Wang, Kaiyang Song, Jiashuo Fan, Yibing Wang, Jingdong Xie, and Zhenzhong Zhang. Hard patches mining for masked image modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10375–10385, 2023a.
- Ying Wang, Tim G. J. Rudner, and Andrew Gordon Wilson. Visual explanations of image-text representations via multi-modal information bottleneck attribution. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023b. URL <https://openreview.net/forum?id=ECvtxmVP0x>
- Jing Xiao, Yufan Bai, Alan Yuille, and Zongwei Zhou. Delving into masked autoencoders for multi-label thorax disease classification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3588–3600, 2023.
- Shuai Zhang, Yiqiu Xu, Naoya Usuyama, Hongfang Xu, Jayesh Bagga, Rema Tinn, et al. Biomed-clip: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023.
- Yiwu Zhong, Jianwei Yang, Pengchuan Zhang, Chunyuan Li, Noel Codella, Liunian Harold Li, Luowei Zhou, Xiyang Dai, Lu Yuan, Yin Li, and Jianfeng Gao. Regionclip: Region-based language-image pretraining. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16772–16782, 2022. doi: 10.1109/CVPR52688.2022.01629.

A USE OF LARGE LANGUAGE MODELS (LLMs)

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